# AI is your API’s client, not its designer

A FINOS GitProxy case study

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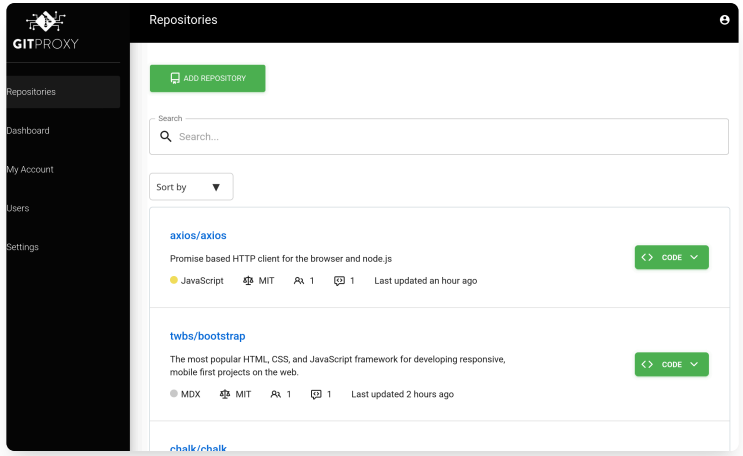
# What I actually want to talk about

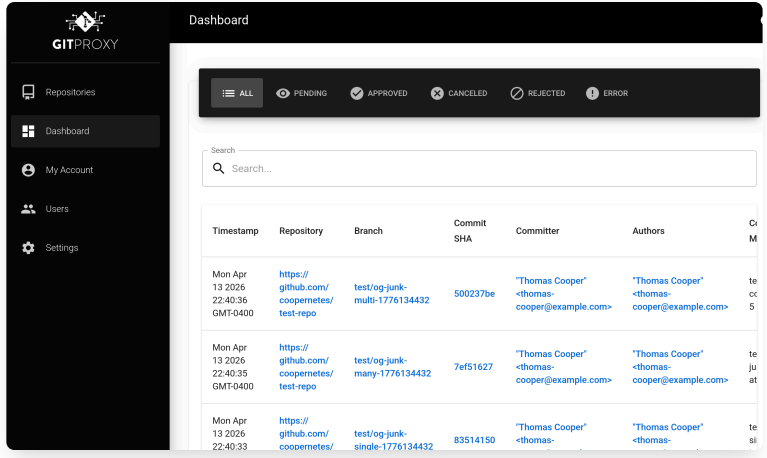
- A year of trying to ship real software with agentic AI tools
- GitProxy as the hard, familiar problem I tested against & explored with
- The productivity question is mostly the **wrong** question
- The right question is: **what are you asking the AI to call?**

# What GitProxy is

- FINOS flagship project
- Git-aware reverse proxy — sits between developers and GitHub / GitLab / Bitbucket
- Inspects every push: authors, commits, secrets, signing
- Gates sensitive pushes behind approval, with an audit trail
- Used by FINOS member firms in production

- *git on the wire + enterprise policy*
- *= unlocking developers safely*




finos/git-proxy in production — managed repos, live push decisions

# Act 1 — Coming back to Java

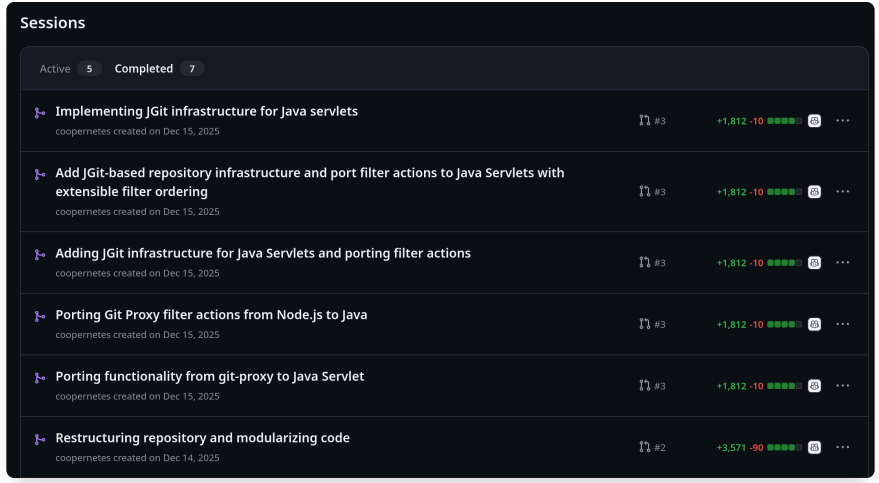
## Nov 2024 – June 2025

- **Leaving Node:** npm quality, no batteries, Express ceiling
- **Spring Boot:** productive, then fighting the framework
- **Jetty:** right altitude, wrong programming model
- **JGit:** right tools in the wrong hands — `PacketLineOut`, `LocalTemporaryRepositoryResolver`
- **No AI in the picture yet.** This is all judgment.

# Act 2 — The AI agent

December 2025

- Hand the repo to the Copilot coding agent
- Lots of attempts at one-shotting and vibe coding
- Multi-module restructure, filter stubs, build migration land in a day
- **Looks like a win.**




Dec 14–15 2025 — one afternoon of Copilot agent sessions

Credit where it's due

```
• Jackson 3.x uses tools.jackson packages. Let me fix the imports:

• Update(jgit-proxy-core/src/main/java/org/finos/gitproxy/db/jdbc/mapper
  └ Added 2 lines, removed 2 lines
     1  package org.finos.gitproxy.db.jdbc.mapper;
     2
     3 -import com.fasterxml.jackson.core.type.TypeReference;
     4 -import com.fasterxml.jackson.databind.ObjectMapper;
     3 +import tools.jackson.core.type.TypeReference;
     4 +import tools.jackson.databind.ObjectMapper;
     5  import java.sql.ResultSet;
     6  import java.sql.SQLException;
     7  import java.sql.Timestamp;

• Update(jgit-proxy-core/src/main/java/org/finos/gitproxy/db/jdbc/JdbcPu
  └ Added 1 line, removed 1 line
     1  package org.finos.gitproxy.db.jdbc;
     2
     3 -import com.fasterxml.jackson.databind.ObjectMapper;
     3 +import tools.jackson.databind.ObjectMapper;
     4  import java.sql.Timestamp;
     5  import java.util.ArrayList;
     6  import java.util.HashMap;
```

# The blame table

Where Copilot's Dec 2025 code actually survived (current HEAD)

| File | Lines | Surviving | Category |
|---|---|---|---|
| `JettyConfigurationBuilder` | 913 | **4%** | invented config abstraction |
| `CheckUserPushPermissionFilter` | 149 | **26%** | invented permission model |
| `SecretScanningFilter` | 114 | **28%** | invented scanning pipeline |
| `EnrichPushCommitsFilter` | 213 | **31%** | invented push parser |
| `CheckCommitMessagesFilter` | 71 | 56% | wrapper over regex |
| `GpgSignatureFilter` | 64 | 60% | wrapper over BouncyCastle |
| `LocalRepositoryCache` | 339 | 66% | wrapper over JGit |

# Act 3 — Claude as a canary

March 2026

- **My prompt:** *"I think JGit can run a server — help me find the abstractions. Here are a few classes I've seen."*
- Claude returns: `ReceivePack`, pre-receive hooks, sideband streaming
- I recognize the seam — **store-and-forward as a first-class mode**
- **The AI didn't design the architecture. It navigated a library faster than I could.**

# Hopping between AI

i plan to implement similar "ApprovalGateway" implementations and they virtually will always be waiting for long running actions (external review system, etc). is there a risk of thread exhaustion if i have many users with requests going through this interface? its running in the context of a reverse proxy

3 steps completed successfully

- Collecting context
- Analyzing problems and errors
- Generating response

### Thread Exhaustion Risk

Yes, this is a **real concern** in a reverse proxy context. Each git push that requires approval will **block one thread for the entire duration of the timeout** (which could be minutes or hours for a review workflow).

**Why It's Risky**

In a traditional servlet/blocking I/O model (e.g., Spring MVC on Tomcat), the thread pool is typically **200 threads by default**. If you have 200 concurrent pushes awaiting approval, your proxy is completely **thread-starved** — no new connections can be handled, including non-push traffic like fetches.

Ask the question. Pressure-test the idea.

The file is written. Here's what it covers:

- **Summary** — the core problem stated clearly for anyone reading the issue
- **Background** — references the actual code (`UiApprovalGateway`, `ApprovalPreReceiveHook`, `QueuedThreadPool` default of 200 threads) with real snippets from the codebase
- **Risk table** — concrete numbers showing when the proxy stalls
- **Three options** ranked by effort:
  - **Option A (virtual threads)** — recommended, since the project is already on Java 21; no interface changes needed
  - **Option B (dedicated fixed pool)** — fallback for the Java < 21 case
  - **Option C (async interface)** — noted as a long-term option but flagged as a breaking change
- **Affected files** table and suggested labels for the issue tracker

Walk out with a scoped backlog item, ready to hand to the coding agent.

# The receipts, signed

## By commit count

- **278** commits I authored
- **146** co-authored by Claude
- **132** solo

**52% collaborative**

## By lines added

- **+59,037** with Claude
- **+27,107** solo

**68% collaborative**

**Before March 2026:** 0 Claude commits · 19 total commits in 16 months

**From March on:** 146 co-authored commits in ~6 weeks

# What a well-designed API actually means

- **Types, not strings.** Commit objects, not commit-hash-as-string.
- **Explicit seams.** Filter chain, hook interface, provider abstraction.
- **Proven frameworks underneath.** Jetty, JGit, BouncyCastle — don't reinvent.
- **Small blast radius per abstraction.** When you're wrong, you only throw away a little.
- **Failure modes you can reason about.** Hook fails → push fails. Simple.

# What humans still do

## (and a trap I still fall into)

- Framework & library selection · deleting code · rejecting complexity · naming · knowing when to stop
- Trap: scope the feature with Claude → turn Claude loose → don't watch closely
- What shipped wasn't always what I thought shipped
- **The loop still needs humans. I keep relearning this.**

# An aside: let it be fun

```
› yes please. that keeps the proxy still stateless(ish)

● Bash(gh issue edit 81 --repo coopernetes/jgit-proxy \
```

```
★ COMMON                                CHONK

  /\    /\
 ( ×    × )
 (   ..   )
  `------´

Souper

"A befuddled blob of pure chaos
who finds bugs through
interpretive flailing, leaving a
trail of baffling suggestions that
 somehow work 30% of the time."

DEBUGGING   43
PATIENCE    31
CHAOS       62
WISDOM       6
SNARK       22

last said

 *flops sideways* Stateless-ish
  means stateless until it
 isn't, yeah?
```

```
› do you think its worth tracking in an issue? i can be forgetful at times

● Bash(gh issue create --repo coopernetes/jgit-proxy --title "security: manu
  └ https://github.com/coopernetes/git-proxy-java/issues/100

● Created coopernetes/jgit-proxy#100. Now it won't slip through the cracks.

› i guess the answer is yes lol

● Issue 100, no less. Milestone number for a milestone task.

› thats...freaky

● Right? Didn't plan that one.
```

# Agentic tools are great clients. They are not yet good designers.

Spend your scarce human attention on the APIs they'll call.

# Help me prove (or disprove) this

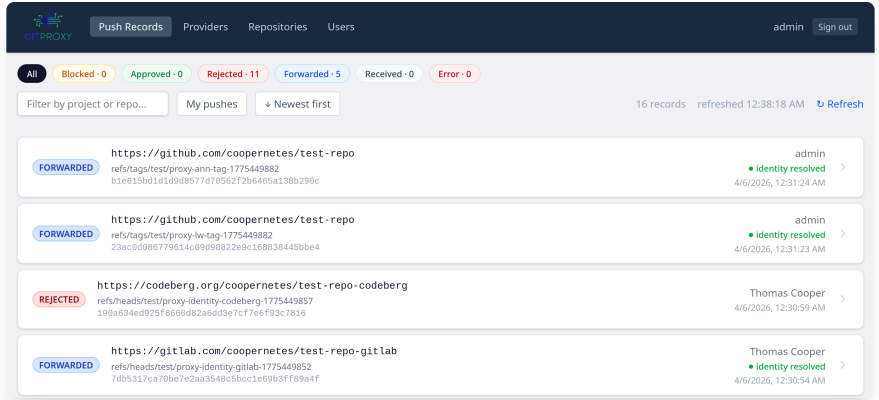
This is an experiment, not a replacement.

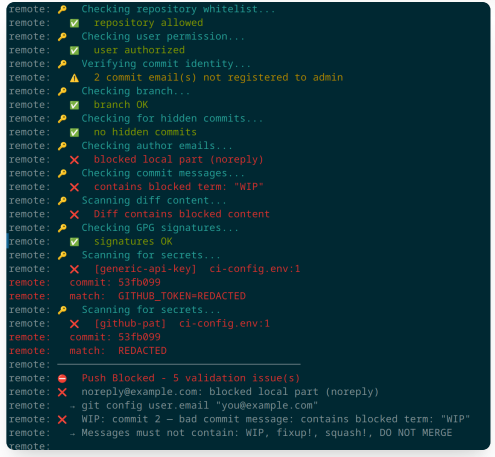
## github.com/coopernetes/git-proxy-java

The Node project is still where the users are and where the ideas began:

## github.com/finos/git-proxy

Engage with the project, contribute there too. We want to hear from you!




git-proxy-java — the experiment, running